A. S. M. SAMIUL ISLAM

Shahid Smrity Hall, BUET, Dhaka, Bangladesh
asm.samiislam@gmail.com | +880-1763-747425
LinkedIn | GitHub

Profile

M.Sc. candidate in the Management of Technology (BUET) with a background in computer science, UI/UX design, and applied AI research. Experience designing digital products, running user research, and building IoT and ML-based assistive technology, with a focus on inclusive and responsible digital innovation. Seeking to contribute to UNDP's Digital, AI and Innovation Hub, particularly in product design and AI tools, research and knowledge management, and programme delivery.

Research Interests

Inclusive Digital Innovation • Human-Computer Interaction (HCI) • User Experience Research • Assistive Technology • Embedded Machine Learning • IoT and Wearable Sensing • Deep Learning • User-Centered Design

Education

Master of Science in Management of Technology (MOT) October 2024 – Present
Institute of Appropriate Technology (IAT),
Bangladesh University of Engineering and Technology (BUET), Dhaka, Bangladesh
• CGPA: 3.67/4.00 | Thesis defence pending
• Thesis: IoT and LLM-based Bangla Sign Language (BdSL) recognition using a wearable smart glove (proposal CASR-approved)
• Supervisor: Tahsina Farah Sanam, Ph.D.

Bachelor of Science in Computer Science and Engineering 2019 – 2023
North South University, Dhaka, Bangladesh
• CGPA: 3.59/4.00 | 50% Merit Scholarship | Distinction: **CUM LAUDE**

Higher Secondary Certificate (HSC), Science July 2016 – June 2018
Govt. Azizul Haque College, Bogura, Bangladesh | GPA: 5.00/5.00

Secondary School Certificate (SSC), Science February 2011 – June 2016
Bogra Zilla School, Bogura, Bangladesh | GPA: 5.00/5.00

Professional Experience

Research Assistant (Part-time) January 2026 – Present
Research and Innovation Centre for Science and Engineering (RISE), BUET, Dhaka, Bangladesh
• Project: "From Rash to Remedy: Tackling Scabies with a One-Stop Digital Healthcare Solution" (ScabieSense); PI: Tahsina Farah Sanam, Ph.D.
• Developing an AI-assisted tool for scabies detection, treatment guidance, and outbreak mapping, now used by doctors screening madrasa students

Postgraduate Research Fellow November 2024 – May 2026
Institute of Appropriate Technology, BUET, Dhaka, Bangladesh
• Developed an IoT and LLM-based Bangla Sign Language recognition system on a wearable sensor glove
• Built ESP32-based sensing hardware with multiple MPU-6050 gyroscopes on a TCA9548A I2C multiplexer

UI/UX Designer and Researcher August 2023 – April 2024
Seed of Intelligence, Mirpur DOHS, Dhaka (Korea-operated)
• Designed AI-powered CCTV-based Video Software as a Service (VSaaS) platform
• Created user flows, schema diagrams, and game UI integrated with VSaaS system
• Designed user diagrams for connected online chat software

Teaching Assistant, Department of Mathematics and Physics June 2021 – January 2022
North South University, Dhaka, Bangladesh
• Assisted five professors in course delivery and student assessment
• Provided academic consultation to undergraduate students

Achievements

• **Creative Visionary** (Certificate of Achievement), Shahid Smrity Hall, BUET, for end-to-end design contributions to the Intra Hall Games 2026 (August 2026)
• **Incubator Selection**, ICSETEP Project, Department of CSE, BUET (August 2025)
• **UIHP Pre-seed Grant**, Team Arora, Cohort IC5, University Innovation Hub Program at BUET (2024)
• **CUM LAUDE Distinction**, North South University (2022)
• **50% Merit Scholarship**, North South University (2019–2022)
• **17th Position**, Regional 3rd Bangladesh Physics Olympiad
• **Divisional 2nd Runner Up**, 9th and 10th Bangladesh Math Olympiad
• **First Position (Team)**, Acting Competition, BTV Subarna Jayanti

Research Experience

IoT and LLM-Based Bangla Sign Language Recognition with a Smart Glove Ongoing
M.Sc. Thesis, Institute of Appropriate Technology, BUET
• Developing a wearable sensor glove and IoT pipeline for BdSL recognition, supervised by Tahsina Farah Sanam, Ph.D.
• Integrating LLMs for sign-to-text and speech output; proposal approved by CASR
• Redesigned my earlier flex-sensor sign-language system around an accelerometer/gyroscope (MPU-6050) setup that gave better results; built on comfortable gaming gloves that fit any hand size, with user testing planned
• Building embedded ML gesture recognition for multimodal communication for deaf and mute individuals

SMMNet-FedHE: Privacy-Preserving Framework for Stereotypical Motor Movement Monitoring with WiFi CSI Accepted, IEEE TENCON 2026
Postgraduate Research, with A. Islam and T. F. Sanam
• Combined federated learning and homomorphic encryption for privacy-preserving, contactless detection of stereotypical motor movements for early autism screening
• Extracted multi-domain features from WiFi CSI data; accuracy above 80% across centralized and federated settings

Wi-SMM: Stereotypical Motor Movement Detection for Autism Screening Using WiFi CSI Submitted
Postgraduate Research, with M. Mohtasim, A. Islam, T. F. Sanam
• Built a WiFi CSI-based dataset and detection framework for early, contactless ASD screening

Wi-Fi CSI-Based Human Activity Detection Ongoing
Independent Project
• Capturing Wi-Fi channel state information (CSI) with two ESP32 boards for activity classification
• Developing the capture firmware with ESP-IDF on macOS

ScabieSense: Scabies Detection, Treatment Guidance, and Outbreak Mapping Ongoing
RISE-BUET, with the Bangladesh Academy of Dermatology (BAD); supported by the International League of Dermatology and BAD
• Researched the screening, case tracking, and follow-up workflow; designed the clinical-AI pipeline (questionnaire scoring, image analysis, fallback logic)
• Focused on mobile-friendly user experience and secure role-based access

BD Pedestrian and Overtaking Action Recognition for Driver Assistance
Undergraduate Research Project, North South University
• Collected and annotated 10+ hours of video dataset for deep learning models
• Developed computer vision algorithms for real-time action recognition
• Improved recognition accuracy for driver assistance systems in Bangladesh

Hakuna Matata – Mental Health Social Network Platform
Undergraduate junior Research Project, North South University
• Designed user-centered interfaces using HTML, CSS, and JavaScript
• Conducted user research on mental health community engagement
• Applied HCI principles to create accessible digital environment

Publications and Manuscripts

• SMMNet-FedHE: Privacy-Preserving Deep Learning-Based Framework for Stereotypical Motor Movement Monitoring with WiFi CSI Data. A. Islam, A. S. M. S. Islam, T. F. Sanam. *Accepted, IEEE*

TENCON 2026.

• Wi-SMM: Detection of Stereotypical Motor Movements for Early Screening of Autism Spectrum Disorder Using a WiFi CSI-Based Dataset. M. Mohtasim, A. Islam, A. S. M. S. Islam, T. F. Sanam. *Submitted.*

Service

University Service

Vice President, NSU HR Club, North South University March 2022 – October 2023

• Organized "HR SPEAKS OUT 4.0" seminar with 700+ attendees
• Led IT, promotional, and strategic planning activities
• Managed recruitment of 300+ members

Sub-Executive (November 2020 – March 2022) | *Senior Member* (March 2020 – November 2020)

Community Service

Panel Deputy Speaker, Child Parliament of NCTF, Bangladesh May 2016 – June 2018

• Prepared speakers for presentations to government ministers
• Facilitated pitches to Education Minister and Deputy Speaker of National Parliament

Trainer and UDRT Member, Bangladesh Red Crescent Society, Bogra Unit March 2015 – August 2016

• Completed disaster management training by Canadian Red Cross
• Led disaster relief operations in Sariakandi, Bogura

Unit Leader, Bogra Zilla School Red Crescent Unit October 2013 – October 2015

• Led team of 100+ active and 300 regular members
• Coordinated relief aid to 100+ flood-affected families

Languages and Skills

Languages

Bengali: Native | **English:** Professional Proficiency

Technical Skills

Programming: Python, C/C++, Java, JavaScript, Swift
Web Technologies: HTML, CSS, Bootstrap, SQL
Mobile Development: iOS (Swift, Xcode)
Design: UI/UX Design, AutoCAD (2D/3D)
Embedded Systems: ESP32 (ESP-IDF), Arduino Uno, NodeMCU, MPU-6050, TCA9548A, Wi-Fi CSI
Data Science: Google Colab, NumPy, Pandas
Tools: Git, MS Office Suite, Google Workspace, LaTeX

Professional Development

• **Supervised Machine Learning: Regression and Classification**, DeepLearning.AI & Stanford University, Coursera August 2025
• **Google UX Design Professional Certificate**, Google, Coursera (UX foundations, wireframing, prototyping, Figma)
• **The Complete 2021 Web Development Bootcamp**, Udemy September 2021

References

Tahsina Farah Sanam, Ph.D.
Professor
Institute of Appropriate Technology
BUET, Dhaka-1000, Bangladesh
Email: tahsina@iat.buet.ac.bd
Phone: +880-1743-489912

Quazi Tafsirul Islam
Senior Lecturer
Department of Management
North South University
Email: quazi.islam@northsouth.edu
Phone: +880-1717-756454